Algebra 2 Name_____

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Quadratic Word Problem WS

1) A fireworks rocket is launched from a hill above a lake. The rocket will fall into the lake after exploding at its maximum height. The rocket's height above the surface of the lake is given by $g(x)= -16x^2 + 64x + 50$. What is the maximum height of the rocket? When does that maximum height occur?

2) A fireworks rocket is launched from a hill above a lake. The rocket will fall into the lake after exploding at its maximum height. The rocket's height above the surface of the lake is given by $g(x)= -16x^2 + 64x + 50$. How long will it take the rocket to hit the lake?

3) A rock is thrown from the top of a tall building. The distance, in feet, between the rock and the ground x seconds after it is thrown is given by $f(x) = -16x^2 - 4x + 125$ How long after the rock is thrown is it 70 feet from the ground?

4) A rock is thrown from the top of a tall building. The distance, in feet, between the rock and the ground x seconds after it is thrown is given by $f(x) = -16x^2 - 4x + 125$ How long after the rock is thrown does it hit the ground?

5) In an effort to catch a criminal; a superhero is going to leap over a building and take a short cut down the ally. The function $f(x) = -16x^2 + 150x$ gives the superhero's height in feet as a function of time. The building is 425 feet hight. Will the superhero make it over the building?

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6) A ball was kicked off the edge of a platform and modeled by the equation $h(t) = -4.9t^2 + 9t + 18$. How long does it take to hit the ground?

7) You are interested in retrieving a souvenir from a shipwreck located 115 feet below the water. You do not have diving equipment, so your dive is limited by the equation $y = 0.05x^2 - 4x - 38$ Assumming you are able to retrieve your souvenir, how long do you have to hold your breath until you resurface?

Solve each equation with the quadratic formula.

8) $11x^2 - x = -10$

9) $8x^2 + 10 = -2x$

Simplify.

10) $(-8 - 6i) + (-3 + 8i)$

11) $(-6 - 4i) - (-1 + 4i)$

12) $-7(8 - 6i) - 6 \cdot -8i$

13) $(2 + 5i) - (-5 - 7i)$

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